

RAIYAN WASIM

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EDUCATION

Galgotias University

B.Tech in Computer Science and Engineering

Aug 2022 – Aug 2026

CGPA: 8.30

Harrow English School

Senior Secondary (CBSE)

May 2019 – July 2021

77.4%

EXPERIENCE

Centre for Railway Information Systems (CRIS), Ministry of Railways

Trainee (Practical Training) — ADAOR Group

Jan 2026 – Jun 2026

New Delhi

- Worked on railway network analysis and freight capacity assessment as part of the Advanced Data Science and Operations Research (ADAOR) Group.
- Processed and cleaned large multi-source railway datasets containing station information, route details, sectional capacities, and operational attributes across multiple railway zones.
- Developed graph-based representations of railway networks to study route connectivity, alternate paths, and sectional constraints.
- Developed a decision-support tool for alternate route discovery and freight capacity estimation between Origin–Destination (O-D) pairs.

CPDD Galgotias University

Student Placement Coordinator (CSE)

Apr 2025 – June 2026

Greater Noida

- Coordinated placement operations across Galgotias University (GU) and Galgotias College (GCET) for campus drives including Infosys and Coding Ninjas in collaboration with HirePro assessment teams.

PROJECTS

Railway Freight Flow Analysis & Visualization | Python, NetworkX, Folium

Jan–Jun 2026

- Developed an interactive railway network analysis platform to identify alternate routes and estimate freight carrying capacity between Origin–Destination (O-D) pairs.
- Processed and integrated railway datasets spanning 17 zones, 69 divisions, 1,285 stations, and 1,962 railway sections into a unified network model.
- Constructed a weighted railway graph with 1,285 nodes and 1,447 edges using NetworkX for route connectivity, alternate path discovery, and freight capacity analysis.
- Applied shortest-path, K-shortest-path, and max-flow based techniques for route analysis and freight capacity estimation between O-D pairs.
- Built GIS-based visualizations with station search, route highlighting, freight flow analysis, network filtering, and interactive exploration features using Folium, JavaScript, HTML, and CSS.

Drive-to-Chat RAG Pipeline | n8n, Google Gemini, Pinecone

July 2025 – Aug 2025

- Built a retrieval-augmented pipeline to convert Google Drive documents into a conversational knowledge base.
- Implemented document chunking and embedding using Gemini APIs and stored vectors in Pinecone.

Static Hosting Platform (Vercel Clone) | Node.js, AWS, Docker

April 2025 – June 2025

- Developed a static site deployment tool that builds projects from GitHub repositories.
- Used Docker for containerized builds and AWS (ECS, S3) for deployment and hosting.

TECHNICAL SKILLS

Languages: C++, Python

Core CS: Data Structures & Algorithms (Sorting, Stacks, Queues, Trees, Heaps), Graph Fundamentals (BFS, DFS, Minimum Spanning Trees, Shortest Paths), Operating Systems, Database Management System (Normalization up to BCNF, Concurrency Control), Computer Networks, Theory of Computation

Tools & Exposure: Git, Docker, AWS (EC2, S3), Pandas, NetworkX, Jupyter Notebook

Cloud: Cloud Computing Fundamentals, AWS (EC2, S3)